# Acheson G163

Acheson presents this problem (though he does not show the solution). The algebra defeated me for quite a long time, but I finally got it.

We are given only the radii of the small circles, $a, b$ and $c$, and are asked to find $r$, which I have written as $R$.

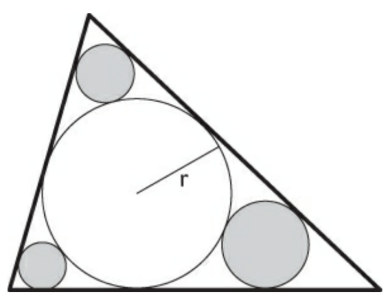


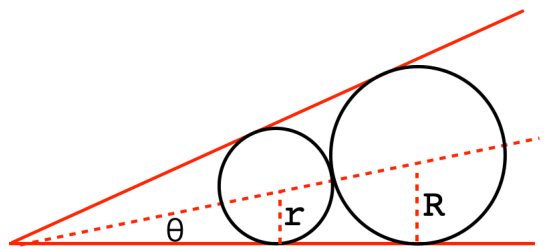

**Fig. 163** A tricky problem from *The Ladies Diary* (1730).

*Solution.*

Let the relevant half-angles be $\alpha, \beta$, and $\gamma$. These are half angles, so they sum to be one right angle. Therefore, any one angle is complementary to the sum of the other two.

$$\sin\alpha = \cos\beta + \gamma = \cos\beta\cos\gamma - \sin\beta\sin\gamma$$

Our previous work with this problem (in the geometry book)




showed that

$$\sin\theta = \frac{R-r}{R+r}, \qquad \cos\theta = \frac{2\sqrt{Rr}}{R+r}$$

Draw similar triangles involving the centers of the two circles to get this result.

We can apply this to the current problem, giving similar formulas for $\alpha$, $\beta$ and $\gamma$ in terms of the radii $a, b$ and $c$ as well as $R$.

So the expression for $\sin\alpha$ above can be written as

$$\frac{R-a}{R+a} = \frac{2\sqrt{Rb}}{R+b} \cdot \frac{2\sqrt{Rc}}{R+c} - \frac{R-b}{R+b} \cdot \frac{R-c}{R+c}$$

Rearranging:

$$(R-a)(R+b)(R+c) + (R+a)(R-b)(R-c) = 4R\sqrt{bc} \cdot (R+a)$$

The first term is

$$\begin{aligned} &(R-a)(R+b)(R+c) \\ &= (R-a)(R^2 + Rb + Rc + bc) \\ &= R^3 + R^2b + R^2c + Rbc - R^2a - Rab - Rac - abc \end{aligned}$$

The second term is

$$\begin{aligned} &(R+a)(R-b)(R-c) \\ &= (R+a)(R^2 - Rb - Rc + bc) \end{aligned}$$

$$= R^3 - R^2b - R^2c + Rbc + R^2a - Rab - Rac + abc$$

The sum has four cancelations and four terms remaining:

$$2R^3 + 2Rbc - 2Rab - 2Rac = 4R\sqrt{bc} \cdot (R + a)$$

We can factor out $2R$

$$R^2 + bc - ab - ac = 2(R + a)\sqrt{bc}$$

$$R^2 + bc - ab - ac = 2R\ \sqrt{bc} + 2a\sqrt{bc}$$

This is a quadratic in $R$:

$$R^2 - 2\sqrt{bc}\ R + bc - ab - ac - 2a\sqrt{bc}$$

The discriminant is

$$4bc - 4(bc - ab - ac - 2a\sqrt{bc})$$

with another cancelation

$$4ab + 4ac + 8a\sqrt{bc}$$

The 4 comes out from the square root as 2 and we have

$$R = \frac{2\sqrt{bc} \pm 2\sqrt{ab + ac + 2a\sqrt{bc}}}{2}$$

$$R = \sqrt{bc} \pm \sqrt{ab + ac + 2a\sqrt{bc}}$$

But what is under the square root is a perfect square, namely

$$ab + ac + 2a\sqrt{bc} = (\sqrt{ab} + \sqrt{ac})^2$$

We have

$$R = \sqrt{bc} \pm (\sqrt{ab} + \sqrt{ac})$$

We will disregard the negative sign and obtain finally

$$R = \sqrt{bc} + \sqrt{ab} + \sqrt{ac}$$

which is the answer we were given. As expected, it is symmetric in $a, b$ and $c$.

$\square$